

# Subagents & Agent Teams Guide

Day 3 Handout - Delegation Patterns (August 2026)

## The Architecture Ladder

| Pattern                  | Use when                                    | Communication                    | Status                     |
|--------------------------|---------------------------------------------|----------------------------------|----------------------------|
| Single session           | One domain, sequential work                 | You Claude                       | Stable default             |
| Subagent                 | Bounded research, review, or verification   | Child reports to parent          | Stable                     |
| Nested subagents         | A coordinator must decompose detailed work  | Results roll up a bounded tree   | Stable, configurable depth |
| Agent team               | Independent peers must communicate directly | Shared task list + peer messages | Experimental               |
| Manual parallel sessions | You want isolated implementation branches   | Human coordinates sessions       | Stable with git worktrees  |

**Default:** use one session. Escalate to subagents when context isolation or independent parallel checks earn it. Use agent teams only when peer communication is central to the task.

## Nested Subagents

Current official behavior:

- Subagents can spawn subagents.
- Default maximum is **three subagent layers below the main conversation**.
- Configure with `CLAUDE_CODE_MAX_SUBAGENT_SPAWN_DEPTH`.
- Set the value to `1` to disable nesting.
- Design shallow: two layers solve most practical decomposition problems.

Example:

```
Main session
Research coordinator
API researcher
database researcher
security researcher
```

Good fit: one coordinator compiles bounded findings from several independent domains. Poor fit: vague "research everything" delegation with no return format or depth limit.

## Background Subagents

As of Claude Code v2.1.198, Claude runs subagents in the background by default unless the agent definition or task requires foreground execution.

Use the **subagent panel** to:

- monitor the tree
- open a transcript
- steer or stop work

- see tool calls and token usage

Background does not mean isolated. Parallel writers in the same checkout can still overwrite one another. Partition file ownership or use an explicit worktree strategy.

## Fan-Out + Grader

---

Use parallel independent candidates when diversity is valuable, then appoint one accountable grader.

```
Main session
Candidate A
Candidate B
Candidate C
Grader -> scored recommendation
```

Require every candidate to return the same format. Require the grader to apply a visible rubric and name a winner.

## Agent Teams

---

Agent teams remain **experimental and disabled by default**.

Enable for a session:

```
export CLAUDE_CODE_EXPERIMENTAL_AGENT_TEAMS=1
claude
```

Agent teams provide:

- a lead agent
- teammate sessions
- a shared task list
- direct peer-to-peer messages

Agent teams do **not** automatically assign separate git worktrees. Teammates usually start in the same working directory. If several teammates will write code, partition file ownership clearly or provide explicit isolation.

### Good team tasks

- independent investigations where peers must challenge or combine findings
- research and review with clear boundaries
- multi-domain work where interfaces genuinely need negotiation

### Poor team tasks

- sequential work
- two teammates editing the same files
- tasks one subagent can complete and summarize
- anything requiring predictable resume, shutdown, or production reliability

## Decision Test

---

Ask in order:

1. Can one session do this safely?
2. Would a bounded subagent keep exploration out of main context?

3. Are there independent shards that can run in parallel?
4. Do peers need to communicate directly while working?
5. Are file ownership, cost, and stop conditions explicit?

If question 4 is "no," prefer subagents. If question 5 is "no," do not parallelize implementation.

## Prompt Pattern - Coordinator + Children

Launch a coordinator subagent to audit the repository.  
 Have it delegate separately to API, data, and security reviewers.  
 Each child must return:

- evidence with file paths
- severity
- one recommended action

The coordinator must deduplicate findings and return one ranked table.

## Prompt Pattern - Fan-Out + Grader

Launch three independent subagents to propose a solution.  
 Each must return: approach, assumptions, risks, and verification plan.  
 Then launch one grader to score each candidate 1-5 for correctness, simplicity, safety, and maintainability. Return the scorecard and winner.

## Cost Controls

- Every subagent has its own context.
- Every agent-team teammate is a separate Claude instance.
- Set a model deliberately for specialist agents.
- Bound the number of candidates, nesting depth, output format, and maximum turns.
- Start the slowest independent task first.
- Stop workers that no longer contribute to the acceptance criteria.

## Production Rule

Subagents are the stable default for delegated reasoning. Agent teams are a supervised experiment, not an automatic production architecture.

Official references:

- <https://code.claude.com/docs/en/sub-agents>
- <https://code.claude.com/docs/en/agent-teams>

(c) 2026 AIA Copilot - Claude Code Training